

Vulnerability Assessment Report

2nd July 2026

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from April 2026 to June 2026. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

The database server of the business houses the main operational data/ information of the business and as such should be guard appropriate access controls in compliance with best practices

The data on the server contains sensitive data of customers and business partners that need to be protected, If not protected could result in a breach leading to leakage of data belonging to customers and necessary for their business operations.

If the data on the server is not properly secured, attacked or disabled it would impact the availability of the services the business renders thereby causing the customers and partner to begin to sort for alternatives.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
An employee	Exfiltration of sensitive company data or altering records, making the attack look like it came from an external hacker rather than an inside employee. Because the server is open to the public.	2	3	3
Hacker	Conduct reconnaissance and surveillance on the business operation and data movements.	3	3	9
A competitor	They simply connect directly to the public IP address, query the database, and silently scrape or download proprietary information, such as client lists, pricing models, or business strategies.	2	3	6

Approach

This evaluation analyzed the organization's data handling and storage architecture. By measuring both the likelihood and severity of potential threats, we ensured that security considerations were balanced with the necessity of maintaining seamless daily business continuity.

Remediation Strategy

The immediate remediation strategy is to block public access by reconfiguring the network firewall or cloud security groups to deny all inbound internet traffic to the database port. The database server must be relocated to a private, internal network segment, completely isolating it from the public internet. If remote administration is necessary, access should be strictly routed through a secure Virtual Private Network (VPN) or a dedicated Bastion host, ensuring the database never possesses a direct public-facing IP address.

To establish defense-in-depth, the organization must enforce strict access controls and continuous monitoring. This includes applying the principle of least privilege through Role-Based Access Control (RBAC), disabling default administrative accounts, and enforcing Multi-Factor Authentication (MFA) for all database access. Additionally, comprehensive audit logging must be enabled and integrated with a Security Information and Event Management (SIEM) system to detect and respond to any unauthorized connection attempts or anomalous queries in real time.